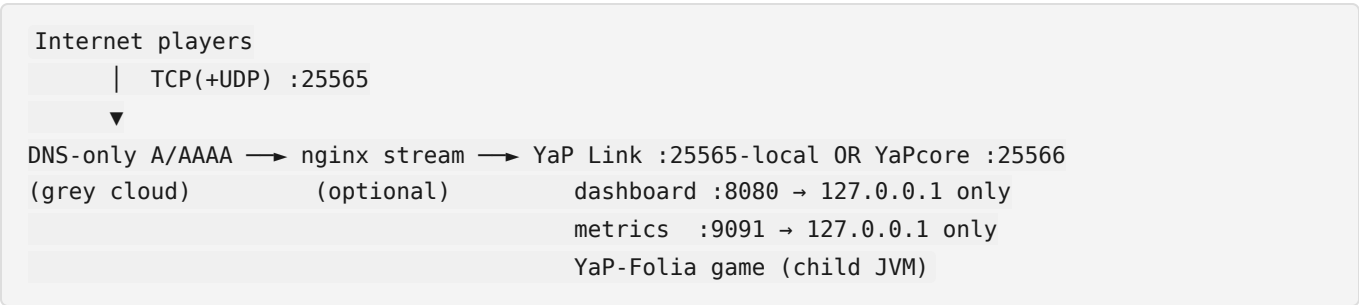


# Edge harden — operator playbook

Harden a **public** YaPcore edge without reading the whole wiki. Builds on [EDGE\\_RATE\\_LIMIT.md](#), [CLOUDFLARE\\_AND\\_NGINX.md](#), and [LAGGUARD.md](#).

## Target architecture



| Surface             | Public?                      | Bind                                       |
|---------------------|------------------------------|--------------------------------------------|
| Game JE/BE          | Yes ( :25565 )               | nginx → Link or YaPcore → <b>YaP-Folia</b> |
| Packs HTTP          | Yes ( :80 / :443 ) via nginx | YaPcore :8081 localhost                    |
| Web dashboard :8080 | <b>No</b>                    | web-dashboard-bind=127.0.0.1               |
| Link /metrics :9091 | <b>No</b>                    | metrics-http-bind=127.0.0.1                |

## Safe defaults

### Public internet

```
# link.properties
connect-rate-limit-enabled=true
connect-rate-per-ip=20
connect-rate-window-ms=10000
handshake-rate-limit-enabled=true
handshake-rate-per-ip=40
login-rate-limit-enabled=true
login-rate-per-ip=10
max-concurrent-per-ip-enabled=true
max-concurrent-per-ip=8
rate-limit-exempt-loopback=true
metrics-http-bind=127.0.0.1
```

```
# config/server.properties
web-dashboard-enabled=true
```

```
web-dashboard-bind=127.0.0.1
web-dashboard-localhost-only=true
internet-exposed=true
```

## LAN / lab

Raise limits or disable concurrent caps if many clients share a NAT:

```
connect-rate-per-ip=100
max-concurrent-per-ip=32
# or max-concurrent-per-ip-enabled=false for LAN only
```

Keep loopback exemption **on** for local admin tools and `./scripts/start.sh`.

## Verify throttles

---

Watch metrics after stressing the edge:

```
yap_link_connect_throttled_total
yap_link_handshake_dropped_total
yap_link_login_dropped_total
yap_link_connect_concurrent_dropped_total
```

Or `GET /api/status` → `observability.linkMetricsHint`.

## nginx stream

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Templates: `deploy/nginx/yapcore-stream.conf.template`

Hardened example (conn limits): `deploy/nginx/yapcore-stream-hardened.conf.example`

```
./scripts/nginx-setup.sh --dry-run
sudo ./scripts/nginx-setup.sh
```

Point **grey-cloud** DNS at the origin. Orange-cloud Minecraft TCP needs Spectrum.

## Cloudflare notes

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1. Game hostname: **DNS only** (grey).
2. Packs/web: orange OK behind nginx `:80 / :443`.
3. Do not publish `:8080`, `:8081`, or Link `:9091` to the world.
4. Optional Spectrum: TCP proxy to origin `:25565` if you must orange-cloud game traffic.

## Fail-closed checklist (under attack)

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1. Confirm drops rising on Link `/metrics` ( `*_throttled` / `*_dropped` ).
2. Tighten: `connect-rate-per-ip=8` , `max-concurrent-per-ip=4` , shorter windows.
3. At nginx: enable `limit_conn` from the hardened stream example; reload nginx.
4. Host firewall: allow only `:25565` (and `:80` / `:443` for packs) from the internet.
5. Temporarily set `internet-exposed=false` / move DNS to maintenance if needed.
6. Do **not** bind dashboard or metrics to `0.0.0.0` to “debug” under fire.

Example host firewall / fail2ban snippets (docs only): see bottom of this file.

## Lag machines

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Install LagGuard ( `gradle installProductDefaults` ). Survival defaults ship in `plugins/YaPLagGuard/config.yml` — [LAGGUARD.md](#).

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## Optional host snippets (examples only)

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### nftables (allow game + packs)

```
table inet yap_edge {
  chain input {
    type filter hook input priority 0; policy drop;
    ct state established,related accept
    iif lo accept
    tcp dport { 25565, 80, 443 } accept
    udp dport 25565 accept
  }
}
```

### fail2ban (jail sketch — tune to your log path)

```
[yap-link-throttle]
enabled = true
filter = yap-link-throttle
logpath = /var/log/yap-link/link.log
maxretry = 1
findtime = 60
bantime = 3600
```

Filter would match `connect throttled ip=` lines when Link logging is raised to INFO for those events. Prefer Link rate limits + nginx `limit_conn` before fail2ban.